9

NAME Lama Alsurakhi

ChE 210 Quiz #3 Tuesday 17 March 2026

To Receive Full Credit, You Must Show All Your Work

Ethanol is converted to acetic acid by the reaction $C_2H_5OH + O_2 \rightarrow CH_3COOH + H_2O$. The process takes place in a batch reactor initially filled with an aqueous solution of ethanol with concentration 1.09 mol/L. A stream of O_2 is passed through the reactor to maintain a constant concentration of dissolved O_2 in the reactor throughout the process. The reaction rate is second order in the ethanol concentration: $r_A = kC_A^2V$ where $C_A = n_A/V$ and V is assumed to be constant at 2500 L. The subscript A indicates the ethyl alcohol. Ignore any volume change caused by the production of water.

(a) If the reaction rate r_A has units of mol/day, and C_A has units of mol/L, what are units for the constant k?

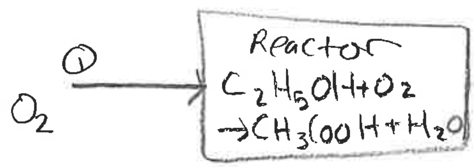
(b) Write a mole balance for ethanol. Combine this with the reaction rate. $\longrightarrow$?. Combine 2 equations = ?

(c) Use the result of part (b) to calculate the time to convert 85% of the ethanol to acetic acid. The numerical value of k is 0.01156 with the appropriate units from part (a).

$C_A(0) = 1.09 \frac{mol}{L}$

$r_A = kC_A^2V$

$V = 2500 L$



a) $r_A = kC_A^2V$ $\quad \frac{mol}{day} = \frac{x}{y}\left(\frac{mol}{L}\right)^2 \cdot L \Rightarrow \frac{mol}{day} = \frac{x}{y}\,\frac{mol^2}{L} \Rightarrow \frac{x}{y} = \frac{L}{day \cdot mol}$

Check: $\frac{\cancel{L}}{day \cdot \cancel{mol}} \cdot \frac{mol^{\cancel{2}\,1}}{\cancel{L^2}} \cdot \cancel{L} = \frac{mol}{day}$

So $\boxed{k = \frac{L}{day \cdot mol}}$

n_A = # moles

remove!

b) $\frac{d\cancel{C}_A n_A}{dt} = \cancel{r_{A,initial}}^{\,0} - r_{A,\text{final or reacted}} \Rightarrow \frac{dC_A}{dt} = -kC_A^2V \Rightarrow \int \frac{dC_A}{C_A^2} = -\int kV\,dt \Rightarrow \int C_A^{-2}\,dC_A = -\int kV\,dt$

ok

$\Rightarrow -\frac{1}{C_A} = -kVt + C \quad , C_A(0) = 1.09 \frac{mol}{L} \Rightarrow -\frac{1}{1.09} = C = -0.917$

$\Rightarrow \boxed{-\frac{1}{C_A} = -2500kt - 0.917}$

other paper $\longrightarrow$

NAME ____________________

Some useful integrals:

$$\int (a+bx)dx = \frac{1}{2b}(a+bx)^2$$

$$\int \frac{dx}{(a+bx)} = \frac{1}{b}\ln|a+bx|$$

$$\int \frac{dx}{x(a+bx)} = -\frac{1}{a}\ln\left|\frac{a+bx}{x}\right|$$

$$\int \frac{x\,dx}{(a+bx)} = \frac{1}{b^2}(a+bx-a\ln|a+bx|)$$

$$\int \frac{dx}{(a+bx)^2} = \frac{-1}{b(a+bx)}$$

$$\int \frac{(A+Bx)}{(a+bx)}dx = \frac{(bA-aB)\ln|a+bx|+bBx}{b^2}$$

$$\int \frac{dx}{(a+bx)(A+Bx)} = \frac{1}{(bA-aB)}\ln\left|\frac{a+bx}{A+Bx}\right|$$

NAME Lama Alsurakhi

$$f_{\text{Conversion}} = \frac{\text{mol init.} - \text{mol}_{\text{final}}}{\text{mol initi.}} = \frac{\text{reacted}}{\text{Feed (Initial)}} \times 100 = 85\%$$

$$\Rightarrow \frac{1.09 - X}{1.09} = 0.85 \Rightarrow 1.09 - X = 0.9265 \Rightarrow X = 0.1635$$

OK

$$-\frac{1}{0.1635}\frac{L}{mol} = -(2500\,L)\left(0.01156\,\frac{L}{day \cdot mol}\right)t - 0.917$$

$$-6.11621\,\frac{L}{mol} = -28.9\,\frac{L^2}{day \cdot mol}\,t - 0.917$$

$$-5.199\,\frac{L}{mol} = -28.9\,\frac{L^2}{day \cdot mol}\,t$$

$$\boxed{t = 0.1799\,\frac{day}{L}} \times 2500\,L = 450 \text{ days}$$

units were trying to tell you "there's a problem here."

$$C_A^{-2} \Rightarrow \frac{-1}{1} C_A^{-1} = -\frac{1}{C_A}$$